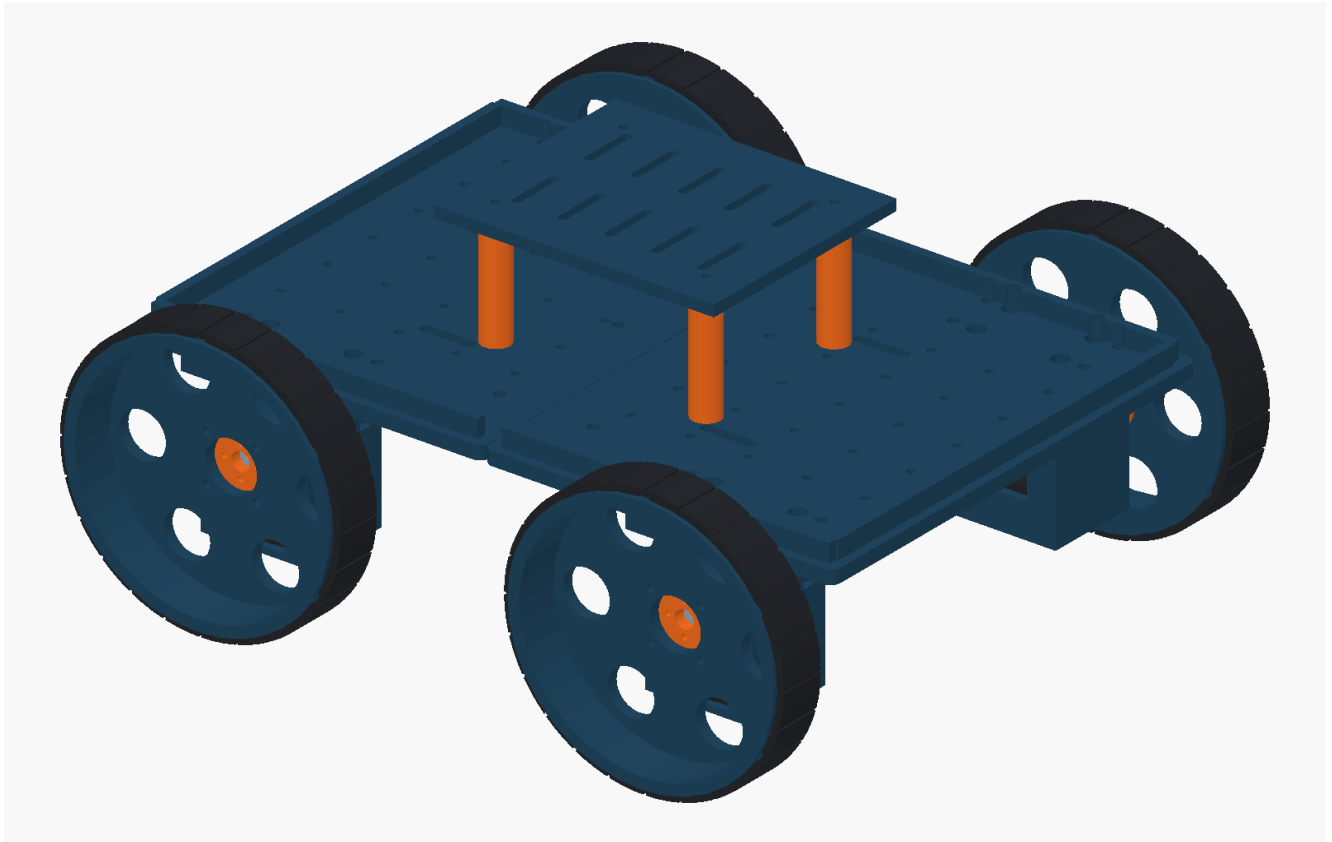


HX-30HM / Mobile Robot Base

R1 - FIRST-ARTICLE BUILD PACKAGE

Real manufacturer servo CAD. Printed chassis, clamps, horn adapters, rims and tyres. M3/M4 structural fasteners.

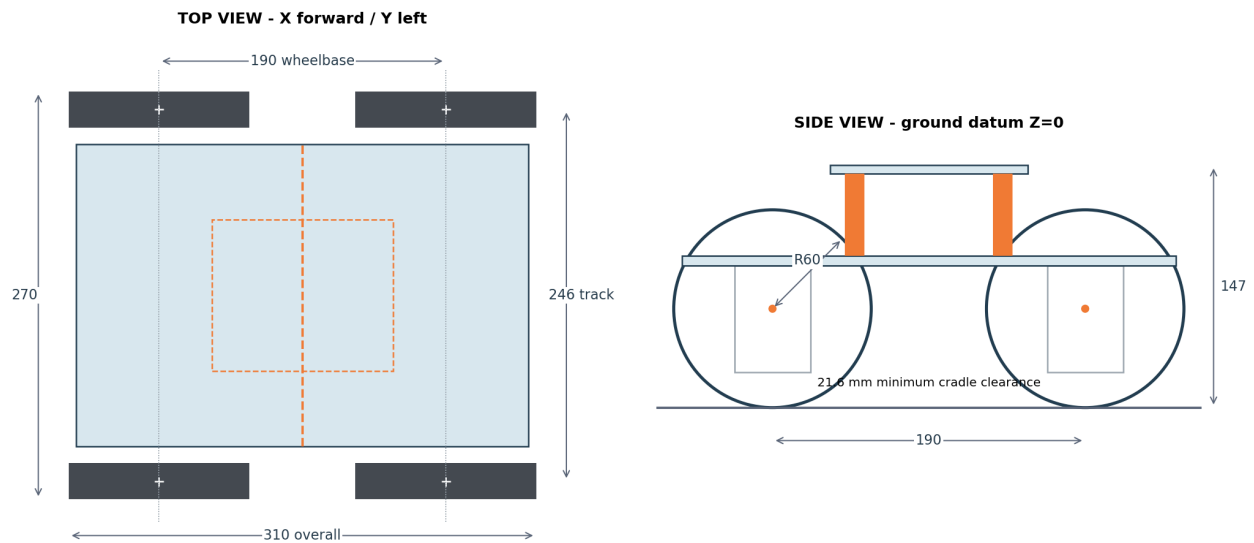


220 x 220 mm print bed | 3 kg total-mass design target | Four-wheel skid steering

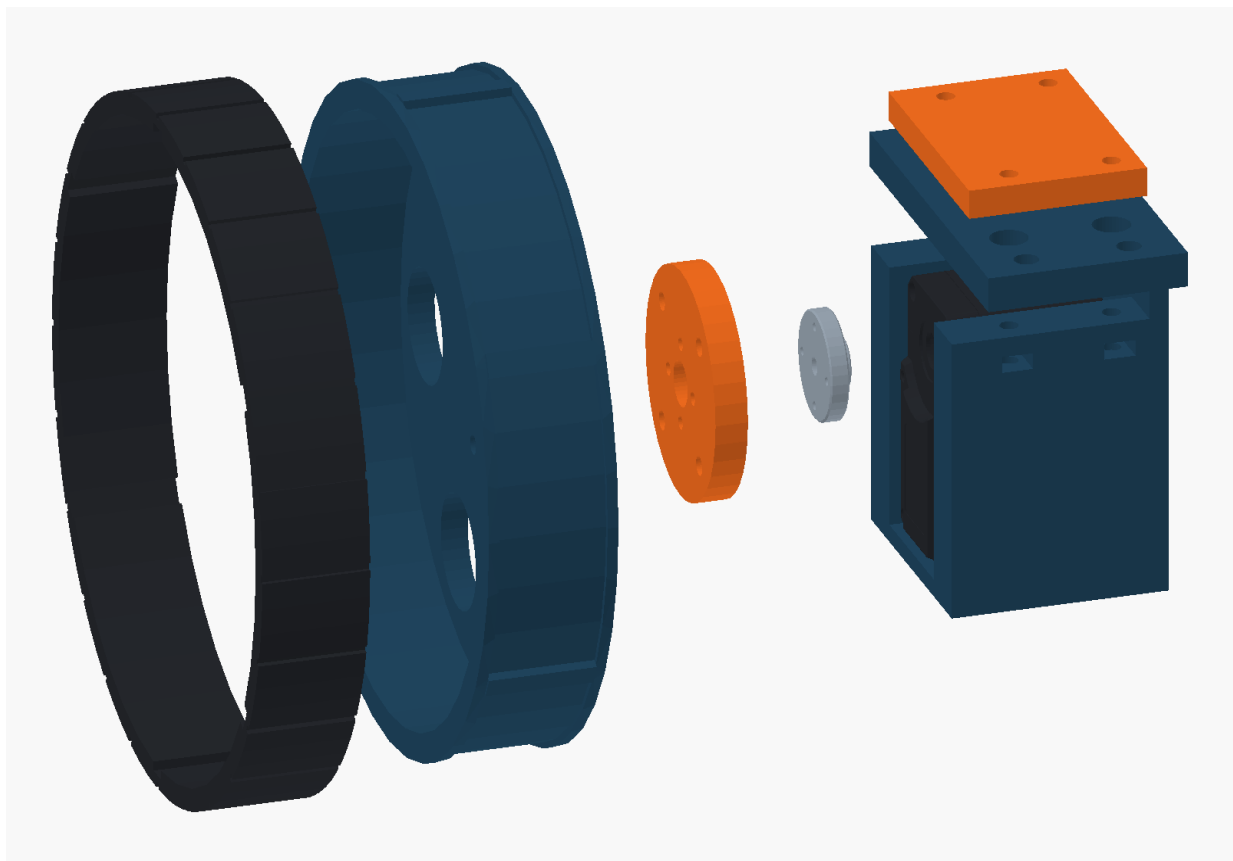
The assembly is geometrically checked, but production release requires physical fit, load and endurance validation. Servo radial wheel-load capacity has not been established.

Preview omits small fasteners for clarity; the STEP assembly includes nominal bolts, nuts and washers. OEM centre retaining screws are specified but not modeled.

Layout and serviceable wheel module



Nominal dimensions in millimetres. Schematic views show design datums; use STEP solids for part geometry.



Exploded view separates the tyre, rim, printed adapter, metal horn, servo cradle and clamp cap. Components share the same axle; separation is illustrative.

Engineering prototype / first-article build. Not released for production. Designed for a 220 x 220 mm print bed and a target total robot mass of 3 kg, including chassis, battery, electronics and payload. This is a design target, not a verified load rating. Dimensions are millimetres.

What changed

This is a new mechanical design, not a rescaled version of the input CAD. The original placeholder motor and damaged wheel have been replaced with manufacturer HX-30HM geometry and a new wheel/horn interface. The chassis has two reinforced halves, three bolted splice plates, four removable servo cradles, separate clamp caps, interchangeable horn adapters, keyed wheel rims, TPU tyres, and a raised universal sensor platform.

Principal dimensions and datums

Item	Dimension
Chassis deck	300 x 200 x 6; upper ribs extend 8 above deck
Main deck underside / top	Z = 86 / 92
Tyre outside diameter / contact width	120 / 20
Rim overall width	24
Wheelbase	190
Track, tyre centre planes	246
Overall wheel envelope	310 long x 270 wide
Ground clearance, lowest cradle	21.6
Sensor platform	120 x 100 x 5; top Z = 147
Wheel centres	X = +/-95, Y = +/-123, Z = 60
Servo output face datums	X = +/-95, Y = +/-100, Z = 60

The world origin is on the floor at chassis centre. X is forward, Y left, Z up. STEP components use functional local datums; the named assembly places them correctly. STL files are independently oriented and translated onto Z=0 for printing; do not assemble using STL origins. CAD/STL use mm; ROS meshes need conversion to metres. STEP export contains named components, not a Fusion parametric timeline or motion joints. Editable CadQuery construction source is supplied in `source/build_robot.py`. Dimensions are defined in that script; `source/nominal_dimensions.json` is a reference summary, not an input configuration file. Install CadQuery 2.x and run `python source/build_robot.py` from the unzipped package; the vendor models must remain in `reference/`.

Printed parts

ID	Part	Quantity	Material
P01	Front deck	1	PETG
P02	Rear deck	1	PETG
P03	Servo cradle	4	PETG
P04	Servo clamp cap	4	PETG
P05	Servo shim	8	TPU
P06	Chassis splice	3	PETG
P07	Horn adapter	4	PETG
P08	Wheel rim	4	PETG

ID	Part	Quantity	Material
P09	Keyed tyre	4	TPU 95A
P10	Sensor platform	1	PETG
P11	Platform spacer	4	PETG
P12	Fastener fit coupon	1 initially	PETG

Use supplied print orientation. Starting settings: 0.4 mm nozzle, 0.2 mm layers, six walls, six top/bottom layers, 40% infill. Use 100% infill for horn adapters and clamp caps. TPU tyres: start with four walls and 25-35% infill; adjust compliance after testing. These are initial process settings, not validated material properties. Print thin shims as solid single layers at their actual 0.25 mm model thickness. Cradles print on their rear face; use local support only if needed around horizontal access holes. Keep support off servo contact surfaces. Splice plates print with nut-pocket face down and may need a small brim. Clear pockets before assembly.

No printed plastic threads carry structural fasteners. M3 holes are 3.4 mm; M4 holes are 4.5 mm. Modelled nut pocket widths are 5.8 mm for M3 and 7.3 mm for M4. Test the coupon: actual clearance depends on printer, material and slicer. Drill/ream clearance holes to the intended diameter if required; do not force bolts through an undersized hole and split a printed part.

Purchased mechanical hardware

Item	Qty	Use
Hiwonder HX-30HM servo	4	One per wheel
Matching metal drive horn, splined side	4	Use the manufacturer's metal spline; do not print it
OEM centre retaining screw	4	Supplied/approved for the purchased servo and horn
M4 x 20 socket-head screw	16	Servo cradles to deck
M4 nyloc nut	16	Cradle mounting, accessible beneath bridge
M4 x 16 socket-head screw	6	Chassis seam splices
M4 regular hex nut	6	Captured in splice plates
M4 flat washer, nominal 0.8 thick	38	32 on cradle joints; 6 above seam joints
M3 x 12 socket-head screw	32	16 clamp caps, 16 rim-to-adapter joints
M3 regular hex nut	32	Captured in cradles and adapters
M3 x 70 socket-head screw	4	Through platform, spacers and main deck
M3 nyloc nut	4	Platform underside
M3 flat washer, nominal 0.5 thick	40	Caps/rim plus both ends of platform joints
M2 x 8 machine screw	16	Adapter to metal horn; verify actual horn thread/depth
M2 flat washer, nominal 0.3 thick	16	Adapter screw heads
20 mm wide reusable battery strap, about 300 mm long	2	Slots in deck; choose length for actual battery

Metric coarse pitches: M2 x 0.4, M3 x 0.5, M4 x 0.7. Socket-head screws: ISO 4762 / DIN 912 type; regular hex nuts: ISO 4032 / DIN 934 type. Use washers against printed surfaces. Nyloc heights vary by supplier; verify bolt protrusion before tightening. The OEM centre screw is an exception: use the supplied correct screw, not a guessed length.

M3/M4 cannot replace the horn's small fasteners. The vendor horn STEP has four nominal 1.6 mm holes consistent with an M2 thread core, on a 14 mm bolt circle. The design uses M2 fasteners there. Confirm the actual purchased horn is threaded M2; do not force M2 screws into an unthreaded or different revision horn.

Assembly sequence

1. **Test one module first.** Print P12, one P03 cradle, one P04 cap, one P07 adapter and shim samples. Check actual servo insertion, connector access, horn diameter/bolt circle and fasteners before printing all parts.
2. Fit the six M4 regular nuts into the three splice plates. Align the deck halves with the 0.3 mm centre gap. Fit splices beneath the seam at Y=-65, 0, +65, using six M4 x 16 screws and top washers.
3. Slide four M3 nuts into each cradle's side-entry nut pockets. Add a lower shim and servo with output facing outward. Feed cables through the rear cable window. Add the upper shim and cap; use four M3 x 12 screws and washers. Tighten evenly only until the servo is retained. The housing must not distort. Adjust shim thickness to eliminate rocking.
4. Attach each complete cradle beneath the deck with four M4 x 20 screws, eight washers and four nyloc nuts. The bridge faces seat directly against the deck underside. Cap screws remain reachable through access holes.
5. Fit the genuine splined metal horn and its OEM centre retaining screw. The rear idler horn is not used. The mating spline is supplied hardware.
6. Insert four M3 nuts in the back of each printed adapter. Attach each adapter to its metal horn using four M2 x 8 screws and washers. Confirm screws engage about 2.9 mm in the modeled horn and do not bottom out. The adapter's 19.4 mm locating recess is based on the vendor CAD's 19 mm horn; measure your horn before printing the set (the marketing drawing rounds its diameter to 20 mm). Change the recess if your revision differs.
7. Stretch each TPU tyre over a rim lip, align the six keys, and seat it between the lips. The keyed features transmit torque; tyre stiffness and fit still need a physical test. Do not use a bare PETG rim as the tyre.
8. Attach each rim to its adapter with four M3 x 12 screws and washers. The 22 mm centre access opening leaves the horn screws accessible.
9. Install the four 50 mm spacers and sensor platform using M3 x 70 screws, washers at both ends, and nyloc nuts underneath the main deck.
10. Mount battery low and near the centre. Use the universal grid/platform for electronics. A specific LiDAR, camera or battery footprint has not been assumed; design its adapter after choosing the exact model.

Servo operation and design limits

The HX-30HM has a velocity-controlled motor mode suitable for continuous wheel rotation. Set unique IDs before connecting all four servos. Use the manufacturer's half-duplex UART interface and compatible bus controller. The advertised 30 kgf.cm value is torque, not a 30 kg payload rating. At 0.19 s per 60 degrees the theoretical unloaded speed is about 52.6 rpm, or 0.33 m/s with these tyres; loaded speed will be lower. It is not a continuous-duty torque or speed guarantee.

The mass target is 3 kg TOTAL. At even static load distribution each wheel supports approximately 7.4 N. A wheel centre 23 mm outboard of the servo front datum creates roughly 0.17 N.m bending moment about that datum. The manufacturer's published radial-load limit was not established. Direct wheel support by the servo therefore requires physical verification. For higher loads, impacts or continuous commercial duty, use a bearing-supported axle or a drivetrain with published wheel-load and continuous-duty ratings. This is a material blocker to a production load rating, not solved by STEP validity. Four-wheel skid steering also increases tyre scrub and drivetrain loads during turns; test on the intended floor.

Required release checks

- Measure actual servo and horn revision; check clamp, shaft engagement, connector clearance, bolt circle, thread size and screw depth on one module.

- Print the fit coupon in the final material. Check nut capture, clamp grip, tyre retention and wheel runout; correct tolerances in source if needed.
- Confirm wheels turn freely by hand, cables cannot touch rotating parts, and every fastener is accessible with the platform installed.
- Test unloaded, then incrementally load to the 3 kg target on a level floor. Record current, temperature, tracking, tyre slip and fastener loosening, including repeated starts, stops and skid turns. Set operating limits from those results and the manufacturer's ratings.
- Inspect printed joints for cracking/creep and perform an agreed proof-load and endurance test before production release. No FEA, fatigue qualification, radial-load qualification or physical printing has been performed here.

Sources and traceability

Manufacturer product: <https://www.hiwonder.com/products/hx-30hm> Manufacturer resource page: <https://www.hiwonder.com.cn/store/learn/43.html> Manufacturer HX-30HM resource folder: https://drive.google.com/drive/folders/1xmsYPLw4wU3SoEZABs0Quua_oJLB2IXZ Servo CAD: <https://drive.google.com/file/d/1LS3cizvNKFOIIMLmUDeMT0-VcW8TMGUd/view> Horn/brackets CAD: https://drive.google.com/file/d/1_G9rTeujNCA0BhDIBWFsojOa9U4fKScV/view Bus modes: https://docs.hiwonder.com/projects/BusLinker/en/latest/docs/1_BusLinker_V3.0_Servo_Debugging_Board_User_Manual.html Protocol: https://github.com/Hiwonder-official/hiwonder-servo-sdk/blob/main/docs/protocol_en.md

Downloaded 2026-10-04. Original manufacturer STEP files are retained in reference/. Their geometry is supplied by Hiwonder; the printed parts and assembly placement are newly designed. The public product page contains some mixed HX-10HM/HX-30HM descriptive text; the CAD dimensions and exact HX-30HM protocol are the primary model/interface references here.

Digital verification result

12 unique printed part types reimported as valid single solids; all fit the 220 x 220 mm print bed. Printed-part interference: 0 overlaps above 0.05 cubic millimetres. Servo envelope intersections with cradle, cap and shims: zero. 12 STL meshes passed closed-edge and positive-volume checks.

Limits: checks exclude fastener thread engagement, OEM spline tolerances, wire bend radii, tyre deformation, real material strength and service loads. See validation.json for the machine-readable results.